

Practice Exam Questions

1. Write a clear one-sentence definition of Level I, Level II, and Level III regression as described by Berk (2010).

2. Suppose you have a fitted logit model. Define a first difference using both words and equations. Be precise! Describe how to interpret the first difference.

3. In the Clark and Golder (2006) normal model exercise, you fit the normal model by maximum likelihood using `optim()`. The parameter vector θ stacks the k regression coefficients β (for the k columns in the design matrix X) and then the scalar σ . Which extraction is correct?
 - A. `beta <- theta[1:(k-1)]; sigma <- theta[k]`
 - B. `beta <- theta[1:k]; sigma <- theta[k+1]`
 - C. `beta <- theta[2:(k+1)]; sigma <- theta[1]`
 - D. `beta <- theta; sigma <- 1`
4. *Select all that apply.* Which of the following formulas **exclude** or **omit** an intercept term from the design matrix?
 - A. `~ 1 + x1 + x2`
 - B. `~ -1 + x1 + x2`
 - C. `~ 0 + x1 + x2`
 - D. `~ x1 + x2`
5. Suppose `species` is a factor with 3 categories. By default, `model.matrix(~ species)` creates:
 - A. Three indicator columns, one for each category
 - B. Two indicator columns (treatment coding), with one category as the baseline
 - C. No columns unless you manually expand
 - D. A single numeric column with levels 1, 2, 3
6. Including `poly(x, 2)` in a formula adds columns x and x^2 to the design matrix.
 - A. TRUE
 - B. FALSE
7. Consider the formula `y ~ x * z`. What predictors does this generate in the design matrix?
 - A. Only the product term $x \times z$
 - B. x , z , and their product $x \times z$
8. Suppose `f` is a formula. Which pipeline ensures that X and y are properly constructed on the same rows (e.g., after dropping missing values from variables used in the model)?
 - A. `X <- model.matrix(f, turnout)`
`y <- turnout$vote`

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B. mf <- model.frame(f, data = turnout)
   X  <- model.matrix(f, mf)
   y  <- model.response(mf)

C. X <- model.matrix(f, na.omit(turnout))
   y <- na.omit(turnout)$vote

```

9. When fitting the logit model via `optim()` with design matrix X , what is a correct choice for the starting parameter vector?
- A. `par_start <- rep(0, nrow(X))`
 - B. `par_start <- rep(0, ncol(X))`
 - C. `par_start <- 0`
 - D. `par_start <- rnorm(nrow(X))`
10. In the probit–logit comparison exercise, you learned the “ ≈ 1.6 rule-of-thumb.” The _____ are about 1.6 times larger in _____ models compared to _____ models.
- A. coefficients; logit; probit
 - B. coefficients; probit; logit
 - C. first differences; logit; probit
 - D. first differences; probit; logit
11. *Choose all that apply.* In the Clark and Golder (2006) t -model, you estimated (β, σ, ν) with ML using a location–scale t distribution. Which statements are true?
- A. With small ν the t has heavier tails than the normal, reducing outlier influence.
 - B. As $\nu \rightarrow \infty$, the t approaches the normal as a limiting case.
 - C. In a location–scale t regression, the link for $\mu = X\beta$ is still the identity; no special inverse link is introduced.
 - D. The t model introduces an additional parameter ν ; the normal model is effectively the t model with $\nu = \infty$.
 - E. The t model always yields identical $\hat{\beta}$ to OLS.
 - F. Smaller ν implies lighter tails than the normal.